



**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**Sixth Semester B.Tech (CBCS) Degree Examinations**

May 2025

**BT3001-1 – BIOINFORMATICS**  
 (For BT)

Duration: 3 Hours

Max. Marks: 100

**Note:**

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS**

**20 Marks**

1. **The first protein sequence (amino acids) reported was:**
  - A) Amylase
  - B) Bovine insulin
  - C) Protease
  - D) Trypsine
2. **Genomics differs from 'classical genetics' in that it considers:**
  - A) Complete hereditary material
  - B) Partial hereditary material
  - C) One Gene
  - D) One gene and one product
3. **The early stages of transcriptome annotations began with which libraries?**
  - A) DNA
  - B) rDNA
  - C) tRNA
  - D) cDNA
4. **Proteomics is the large-scale study of:**
  - A) RNA
  - B) DNA
  - C) Proteins
  - D) Carbohydrates
5. **Long text file that contains many entries separated by a delimiter:**
  - A) Relational
  - B) Flat file
  - C) Object oriented
  - D) ER diagram
6. **Databases typically use this to provide access to additional information:**
  - A) ID numbers
  - B) Keywords
  - C) Hyperlinks
  - D) Reference titles
7. **NCBI refers to**
  - A) National Center for Biodiversity Information
  - B) National Center for Biology Information
  - C) National Center for Bioinformatics Information
  - D) National Center for Biotechnology Information
8. **Computationally post processed sequence information is available in:**
  - A) Literature Databases
  - B) Secondary database
  - C) Archival Databases
  - D) Primary Databases
9. **It is the large archive since 1971 for biological macromolecular structures:**
  - A) PDB
  - B) PubChem
  - C) CATH
  - D) SCOP
10. **Identifying similar structures between proteins provide the opportunity to recognize:**
  - A) Homology
  - B) Heterology
  - C) Sequence comparison
  - D) Molecular mechanism
11. **An algorithm and program for sequence alignment developed by NCBI**
  - A) Needleman- Wunsch
  - B) FASTA
  - C) Smith- Waterman
  - D) BLAST
12. **One of the largely acclaimed structure-based approaches in drug discovery:**
  - A) Molecular modelling
  - B) Molecular docking
  - C) Molecular folding
  - D) Molecular recognition

13. A sequence can be plotted against itself and regions that share significant similarities in  
 A) Word method B) Dynamic method  
 C) Dot plot method D) Heuristic method
14. In Position-Specific Scoring Matrices the negative score represents a:  
 A) Non-conserved sequence B) Conserved sequence  
 C) Identical residue D) Similar residue
15. BLOSUM62 means that the sequences clustered in this block are at least:  
 A) 31% similar B) 31% non-identical  
 C) 38% similar D) 62% identical
16. It is a series of widely used computer programs used for multiple sequence alignment:  
 A) T-Coffee B) Clustal  
 C) BLAST D) FASTA
17. A group of taxa descended from a single common ancestor is defined as a:  
 A) Taxa B) Root  
 C) Clade D) Node
18. Which among the following is not a distance-based method of phylogenetic tree construction:  
 A) UPGMA method B) Maximum Parsimony method  
 C) Neighbor Joining method D) Minimum Evolution method
19. Character-based phylogenetic tree building methods are also known as:  
 A) Discrete methods B) Distance methods  
 C) Minimum Evolution methods D) Conservative methods
20. All the bootstrapped trees are summarized into a consensus tree based on a:  
 A) Minority rule B) Majority rule  
 C) Consensus rule D) Confidence rule

### **PART - B: DESCRIPTIVE ANSWER QUESTIONS**

| Unit – I   |                                                                                                      | Marks | BT* | CO* | PO* |
|------------|------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.         | a) Explain goals and outcomes of Human Genome project.                                               | 8     | L*1 | 1   | 1,2 |
|            | b) Describe chain termination method of genome sequencing with neat labelled diagram.                | 8     | L2  | 1   | 2,4 |
| 2.         | a) Outline the whole genome sequencing significance and advantages.                                  | 8     | L2  | 1   | 2,5 |
|            | b) Elaborate on Transcriptomics and Metabolomics.                                                    | 8     | L1  | 1   | 2,5 |
| Unit – II  |                                                                                                      |       |     |     |     |
| 3.         | a) Illustrate Entity Relationship (ER) diagrams with examples.                                       | 8     | L1  | 2   | 3,5 |
|            | b) Write brief description on any two primary databases which are useful for biological researchers. | 8     | L1  | 2   | 3,5 |
| 4.         | a) Secondary biological databases are derived from primary data. Discuss any two of them.            | 8     | L1  | 2   | 3,5 |
|            | b) Explain the principles of primer design along with software tool.                                 | 8     | L2  | 3   | 5   |
| 5.         | a) Give an account on the proteins structural visualization tools with their special features.       | 8     | L1  | 2   | 3,4 |
|            | b) Discuss the importance of computer aided drug design (CADD) for drug discovery.                   | 8     | L2  | 3   | 5   |
| Unit – III |                                                                                                      |       |     |     |     |
| 6.         | a) Describe the Local sequence alignment algorithm with an example.                                  | 8     | L2  | 4   | 2,4 |
|            | b) Summarize PAM and BLOSUM scoring matrices along with their key features.                          | 8     | L1  | 5   | 2,4 |
| 7.         | a) Relate pairwise and multiple sequence alignment with applications.                                | 8     | L2  | 4   | 2,4 |
|            | b) Give an account on any two statistical evolutionary models employed in phylogenetic analysis.     | 8     | L2  | 5   | 5   |
| 8.         | a) Internet based tools such as Clustal, T-Coffee, are available for sequence alignment. Justify.    | 8     | L2  | 4   | 3,5 |
|            | b) Explain the phylogenetic tree construction steps based on the evolutionary relationship.          | 8     | L2  | 5   | 4,5 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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